GENERATIVE AI MINI PROJECT

CYBERSECURITY FUNDAMENTALS

Study Material for an AI Personal Study Assistant

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Student Name	Muhsina S
Department	B.Tech CSE (Artificial Intelligence)
Institution	TKM Institute of Technology, Kollam

A beginner-friendly guide to protecting devices, accounts, networks, and information.

1. Introduction to Cybersecurity

Cybersecurity is the practice of protecting computers, mobile devices, networks, applications, and data from unauthorized access, damage, theft, or disruption. It combines technology, processes, and user awareness to reduce digital risks.

Why is cybersecurity important?

• Protects personal information such as passwords, photos, messages, and banking details.
• Helps organizations keep services available and protect customer data.
• Reduces financial loss, identity theft, and reputational damage.
• Supports privacy, trust, and safe use of online services.

The CIA Triad

Confidentiality: Information is seen only by people who are authorized to access it. Example: using access permissions to keep a student record private.

Integrity: Information remains accurate and is not changed without authorization. Example: protecting marks in a college database from unauthorized editing.

Availability: Systems and information are accessible when authorized users need them. Example: maintaining backups and reliable servers so a learning portal can be used during exams.

Key terms

Asset: Something valuable, such as data or a device. **Threat:** Something that could cause harm. **Vulnerability:** A weakness that can be exploited. **Risk:** The possibility and impact of a threat exploiting a vulnerability. **Control:** A safeguard used to reduce risk.

A simple example

A student uses the same easy password for email and a shopping account. If one service is breached, an attacker may try that password elsewhere. The password is a weak point, account takeover is a threat outcome, and using unique passwords with multi-factor authentication reduces the risk.

2. Common Cyber Threats

Cyber threats use technical weaknesses or human mistakes to steal information, interrupt services, or gain unauthorized access. Learning the warning signs helps users respond before damage occurs.

Malware

Malware means malicious software. Common types include:
• **Virus:** Attaches to a file or program and spreads when it runs.
• **Worm:** Can spread between systems without needing a user to run an infected file.
• **Trojan:** Pretends to be legitimate software while carrying harmful functionality.
• **Ransomware:** Locks or encrypts data and demands payment.
• **Spyware:** Secretly collects information about a user or device.

Phishing and social engineering

Phishing is a deceptive message designed to make someone click a harmful link, open an attachment, or reveal information. **Social engineering** is the broader use of manipulation to persuade people to take unsafe actions. Warning signs include urgent demands, unexpected attachments, mismatched sender addresses, unusual payment requests, and links to unfamiliar websites.

Other common threats

• **Password attacks:** Guessing, reusing, or stealing passwords.
• **Denial-of-Service (DoS/DDoS):** Flooding a service with traffic so legitimate users cannot access it.
• **Man-in-the-Middle (MitM):** Intercepting or altering communication between parties.
• **Insider threat:** Harm caused by someone with legitimate access, intentionally or accidentally.
• **Unpatched software:** Attackers exploit known weaknesses that have not been fixed.

Example: spotting a phishing message

A message claims, “Your account will be closed in 10 minutes—verify your password here.” Do not click the link. Open the official app or type the known website address yourself, check the sender, and report the message through the service’s official process.

3. Essential Cybersecurity Practices

Strong cybersecurity depends on everyday habits as well as technical tools. The following practices are useful for students, families, and organizations.

Password security

• Use a long, unique password or passphrase for every important account.
• Use a reputable password manager to create and store passwords.
• Never share one-time passwords (OTPs), recovery codes, or passwords with strangers.
• Turn on **multi-factor authentication (MFA)**, which requires an additional proof of identity beyond the password.

Updates and device protection

• Enable automatic updates for the operating system, browser, and apps.
• Install software only from trusted stores or official websites.
• Use built-in security protections and reputable anti-malware tools where appropriate.
• Lock devices with a PIN, password, or biometrics, and enable device-finding or remote-wipe features if available.

Safe browsing, email, and downloads

• Check the full website address before entering sensitive information.
• Avoid unexpected links, attachments, cracked software, and unknown USB drives.
• Confirm unusual requests for money or confidential information using a separate trusted channel.
• Do not assume a website is safe simply because it looks professional or uses HTTPS; HTTPS encrypts the connection but does not guarantee that the site is honest.

Wi-Fi and backups

• Secure home Wi-Fi with a strong router administrator password and modern encryption such as WPA2-AES or WPA3.
• Avoid sensitive transactions on untrusted public Wi-Fi when possible; use trusted networks and appropriate protection.
• Back up important files regularly. Keep at least one backup separate from the device so malware or device failure is less likely to destroy every copy.
• Test that backups can actually be restored.

Privacy and social media

Limit public sharing of your address, phone number, travel plans, student ID, and other personal details. Review app permissions and account privacy settings. Think carefully before posting photographs that reveal private information.

4. Networks, Encryption, and Access Control

Network security

A computer network connects devices so they can share data and services. Network security aims to prevent unauthorized access, misuse, and disruption.

• **Firewall:** Filters network traffic according to security rules.
• **Antivirus/anti-malware:** Detects, blocks, or removes many forms of malicious software.
• **Virtual Private Network (VPN):** Creates an encrypted tunnel between a device and a VPN endpoint; it does not make every online activity anonymous or automatically safe.
• **Secure Wi-Fi:** Uses appropriate encryption and strong router settings.
• **Network segmentation:** Separates systems into sections to limit the spread of an attack.

Encryption

Encryption transforms readable information (plaintext) into unreadable ciphertext using a cryptographic method and key. Decryption converts it back for an authorized party. Encryption helps protect confidentiality when data is stored or transmitted.

Two broad types

• **Symmetric encryption:** The same secret key is used to encrypt and decrypt data. It is generally efficient for large amounts of data.
• **Asymmetric encryption:** Uses a public-key/private-key pair. It supports tasks such as secure key exchange and digital signatures.

Authentication and authorization

Authentication answers “Who are you?” and verifies identity, for example with a password and fingerprint.
Authorization answers “What are you allowed to do?” and determines which files or actions an authenticated user can access.

Principle of least privilege

Give each user or application only the access needed to do its job. Review permissions regularly and remove access when it is no longer required. This limits the damage caused by compromised accounts.

Example

In a college portal, a student may view their own marks, while an authorized faculty member may enter marks for a class. Logging in proves identity; the portal’s permissions determine what each person can do.

5. What to Do During a Cybersecurity Incident

A cybersecurity incident is an event that may compromise the confidentiality, integrity, or availability of systems or data. Quick, careful action can reduce harm.

If you suspect an account is compromised

1. Use a trusted device and go directly to the official service website or app.
2. Change the affected password and any reused passwords.
3. Enable MFA and review recovery email addresses, phone numbers, active sessions, and connected apps.
4. Sign out unknown sessions and revoke suspicious access where the service permits it.
5. Contact the service provider, bank, or college IT team if sensitive information or money may be at risk.

If malware is suspected

- Stop entering passwords or sensitive information on the affected device.
- Disconnect it from Wi-Fi or the network if doing so is appropriate and safe.
- Tell the responsible IT/security team; follow their instructions before deleting files or reinstalling software.
- From a clean device, change important passwords if they may have been exposed.
- Restore files from known-good backups only after the underlying issue has been addressed.

If you clicked a suspicious link

Do not panic. Close the page, do not download or run anything else, and do not enter more information. If you entered a password, change it from a trusted device. If you provided bank details or approved a payment, contact the bank immediately through its official number or app. Report the message to the relevant organization.

Incident response cycle

Prepare → Detect and analyse → Contain → Eradicate → Recover → Learn. Preparation includes backups and contact lists. Detection identifies the event. Containment limits its spread. Eradication removes the cause. Recovery safely restores services. Lessons learned improve future protection.

Important

Preserve useful evidence such as the time, sender, screenshots, and alert messages, but do not forward malicious files or expose private information. Follow your institution's reporting procedure. Never attempt to retaliate against or hack the suspected attacker.

6. Cybersecurity for Students

Students use learning portals, email, cloud storage, social media, and payment apps. A few consistent habits can prevent many common incidents.

Student cybersecurity checklist

- Use a unique password for college email and learning platforms.
- Enable MFA wherever available.
- Check the sender and destination before opening links or attachments.
- Keep laptops and phones updated and screen-locked.
- Download lecture notes and software only from trusted sources.
- Back up assignments and project files to a secure location.
- Review privacy settings and avoid sharing student credentials.
- Report suspicious messages to faculty or the IT support team.
- Log out on shared computers and never save passwords on public devices.

Short case study

Situation: A student receives an email that appears to come from the college IT department. It asks the student to open an attachment to “confirm exam registration.” The sender address is unfamiliar and the email creates urgency.

Risk: The attachment may contain malware, or the link may lead to a fake login page.

Safe response: Do not open the attachment. Contact the college through a known official channel, report the email, and delete it if instructed. If it was already opened, tell the IT team promptly and follow their advice.

Quick glossary

Patch: A software update that fixes issues or vulnerabilities.
Backup: A separate copy of data used for recovery.
MFA: Multi-factor authentication.
Data breach: Unauthorized access to or disclosure of information.
Vulnerability: A weakness that could be exploited.
Incident response: Coordinated actions to handle a security event.

7. Revision Questions and Summary

Key points to remember

• Cybersecurity protects systems, networks, applications, and information.
• The CIA Triad stands for Confidentiality, Integrity, and Availability.
• Malware, phishing, password attacks, and unpatched software are common threats.
• Unique passwords, MFA, updates, careful browsing, and tested backups reduce risk.
• Encryption protects data; firewalls filter traffic; access control limits permissions.
• During an incident, report quickly, contain the problem safely, and follow trusted recovery steps.

Practice questions

1. Define cybersecurity and explain why it is important.
2. Explain the three parts of the CIA Triad with examples.
3. What is malware? Describe any four types.
4. What is phishing? List five warning signs of a suspicious message.
5. Explain five good cybersecurity practices for students.
6. Differentiate authentication and authorization.
7. What is encryption? Briefly compare symmetric and asymmetric encryption.
8. What steps should a student take if their account may be compromised?

Suggested answers — short revision

1. Cybersecurity protects digital systems and information from unauthorized access, damage, theft, and disruption.
2. Confidentiality prevents unauthorized viewing; integrity prevents unauthorized changes; availability keeps services accessible.
3. Malware is harmful software. Examples include viruses, worms, Trojans, ransomware, and spyware.
4. Phishing is a deceptive attempt to steal information or trigger unsafe actions. Warning signs include urgency, unknown senders, mismatched links, unexpected attachments, and requests for passwords or OTPs.
5. Use unique passwords, MFA, updates, trusted downloads, and regular backups.
6. Authentication verifies identity; authorization determines permitted actions.
7. Encryption converts plaintext into ciphertext. Symmetric encryption uses one shared secret key; asymmetric encryption uses a public/private key pair.
8. Change the password from a trusted device, enable MFA, review sessions and recovery options, and contact the service provider or IT team.

End of Study Material